

Optimizing Customer Purchases and Sales Forecasting for a Bakery Business

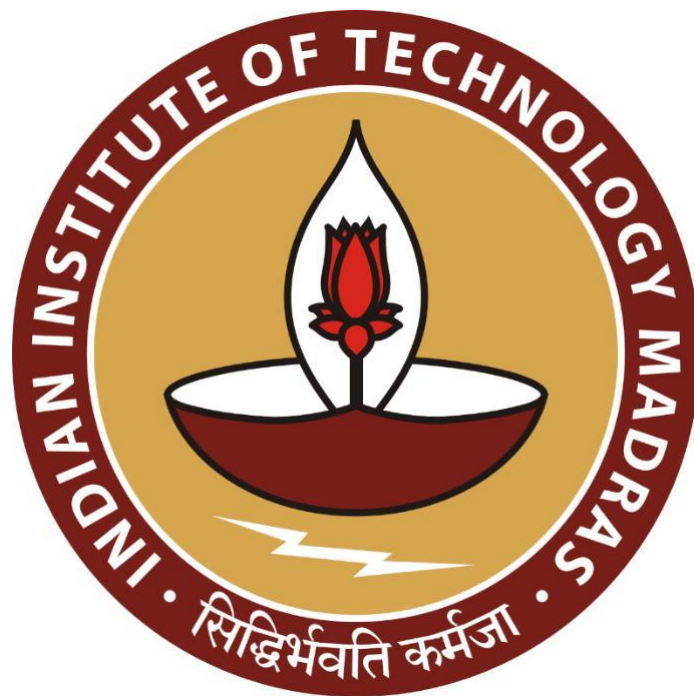
A Final report for BDM capstone project

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Executive Summary

This project analyzes the sales performance at a bakery named “The Bread Basket” situated in Edinburgh, Scotland, which operates in a high-traffic retail environment with the goal of improving business operations and resource planning. Two core business problems were identified: (1) low average basket size, indicating missed opportunities for increasing revenue per transaction and (2) significant variability in daily sales and weekly sales, leading to inefficiencies in staffing, and inventory planning. This project aims to address these issues through data-driven analyses on historical transactional data.

The data used in the project is sourced from a widely popular data science platform Kaggle titled “Bakery Sales”. It comprises over 9,000 transactions from 30-Oct-2016 to 9-Apr-2017. The dataset contains fields (columns) such as Transaction, Item, date_time, period of day (morning/afternoon/evening/night), and Day Type (weekend/weekday). The dataset was cleaned and contained no null values. For problem 1, Market Basket Analysis using Apriori algorithm was conducted, and for problem 2, time series forecasting was applied using SARIMAX and XGBoost, along with Monte Carlo simulation to model uncertainty.

The analysis revealed that Coffee, Bread, and Tea were the top-selling items. High-lift item combinations were identified, indicating bundling and upselling opportunities. Time-based analysis showed strong morning sales and afternoon sales while evenings remained underutilized. Time series models effectively captured both weekly and daily demand trends.

Key recommendations include bundling top-selling items to increase the average basket size, and offering targeted promotions during underutilized hours. The time series models were used to predict the daily sales pattern. Implementing these strategies is expected to enhance customer satisfaction, improve operational efficiency, reduce wastage, and drive long-term financial growth.

Proof of Originality

The dataset used in the project is publicly available on Kaggle, a reputable data science and machine learning platform. The dataset comprises over 9,000 transaction records of a bakery called “Bread Basket” located in Edinburgh, Scotland. The dataset consists of 20,507 entries, with each row containing information about the particular item sold, along with the specific time of the day and day of the week when the transaction occurred. The data covers a time period from 30th October 2016 to 9th April 2017.

The dataset was independently collected and published by the original uploader on Kaggle titled “Bakery Sales”.

Link of the Dataset: <https://www.kaggle.com/datasets/akashdeepkuila/bakery/data>

Meta Data

The dataset named “Bakery Sales” has 20,570 rows, and contains 5 columns that is: Transaction, Item, date_time, period_day, weekend_weekday.

Below is the metadata about the dataset.

Column Name	Data Type	Description	Value Count	Format	Relevance to Project
Transaction	Integer	Unique ID for Each Transaction	9465	Count	It will be used for Market Basket Analysis.
Item	String (Categorical)	Name of the Item sold	94 Unique Items	Product Name	Identifies top sellers and cross-selling combinations
date_time	Timestamp	Date and Time of Transaction	30 Oct 2016 to 9 Apr 2017	DD/MM/YYYY HH:MM	Enables time series forecasting and seasonal analysis
Period_day	String (Categorical)	Time block of day when transaction occurred	Morning/ Afternoon/ Evening/ Night	Nominal Category	Helps analyze sales by time of day
Weekend _ Weekday	String (Categorical)	Type of day when transaction occurred	Weekday/ Weekend	Nominal Category	Identifies weekday vs weekend purchase patterns

Table 1: Metadata of Variables in Bakery Sales Dataset

Descriptive Statistics

The dataset comprises of 20,507 rows with 9,465 transactions occurring between 30-Oct-2016 to 9-Apr-2017, and have 5 columns. It has one Identifier column that is transactions, and 4 categorical columns named: “Item” (Name of Item), “date_time” (date and time of transaction), “period_day” (Time block of Transaction), “Weekend_weekday” (Day type of transaction).

After that some feature engineering has been performed on the dataset to extract month, day, year of transaction, and grouped the transactions on basis of date_time column to check how many items got sold per transaction, and new column is created named “Basket_Size”.

As transaction is a unique Identifier for each row, it will be excluded from the descriptive statistics.

The following table summarizes the distribution of basket size across all transactions.

Column Name	Basket_Size
Count	9465
Mean	2.16
Standard Deviation	1.30
Minimum	1
25 th percentile	1.0
50 th percentile (Median)	2.0
75 th percentile	3.0
Maximum	11

Table 2: Descriptive Statistics of numerical columns

The data shows that most customer purchases 1-2 items per visit. With an average basket size of 2.16, and low standard deviation indicates that customer buy only similar number of items. 50% of the customers buy between 1 and 3 items.

The following table summarizes the descriptive statistics of categorical columns.

Column Name	Count	Unique Values	Most Frequent Value	Frequency of top category
Item	20507	94	Coffee	5471
date_time	20507	9182	2/5/2017 11:58	12
period_day	20507	4	Afternoon	11569
weekend_weekday	20507	2	Weekday	12807

Table 3: Frequency distribution of Categorical Columns

The table 3 shows that categorical variables provides valuable insights with respect to customer behaviour. Coffee is the most sold item, making it strong anchor product for bundling, so coffee will be the centre product for making connections with other product in market basket analysis. Sales are higher in Afternoon, indicating peak operational hours.

Furthermore, the majority of transaction occurs on weekdays, indicating the need of better inventory planning and proper staffing during weekdays.

Detailed Explanation of analysis process/methods

Data Cleaning and Preprocessing

To ensure the data was suitable for analysis, several cleaning and preprocessing steps were carried out. These steps aimed to enhance data cleaning, removing inconsistencies, and generate new features (columns) that were relevant to the business problem identified in the project.

Step 1: Handling Missing Values and Duplicates

- Before conducting Explanatory Data Analysis (EDA), all the features were checked for the missing (null) values and duplicate entries. No missing values or duplicate transactions were found, confirming the completeness and consistency of the dataset.

Step-2: Date-Time parsing:

- The date_time column in the dataset was of object specifically string datatype, and it is converted into python's date_time format using `pd.to_datetime()`. Here `pd` is an alias refers to Pandas library which is used for data analysis.
- This transformation enables time based filtering, resampling, and creating new features based on days and hours.
- This also facilitates time series analysis and time period classification.

Step-3: Feature Engineering:

- A new DataFrame named "df_grouped" was created with the features "Transactions", "Item", "Date", "Day", "Year", "Month", "Basket_Size".
- This is done by grouping the transactions based on the date_time column of the original data. Most importantly, a new feature named Basket_Size was generated by counting the number of Items per unique transaction. The column was essential for analysing the average number of items purchased per visit.
- Another DataFrame with the name "Daily Sales" is created with date_time as index and sales with feature

Analysis Process/ Methods

Problem 1: Low Average Basket Size

To address low basket size problem, there are several different techniques for it but one of the most accurate and popular is market basket analysis. This method is well-suited for discovering frequent-item bought together in the data which directly aligns with objective of the problem that will encourage multi-item purchases. Apriori is specially designed for these kinds of problem, it identifies the relationship between these items, enabling actionable strategies to increase basket size.

Following are the steps taken in doing market basket analysis.

1) Data Transformation

- The Item were extracted from the df_grouped DataFrame, where the column “Item” contains the list of items sold per transaction, Based on this, a list of itemsets named transactions was created to serve as input for TransactionEncoder.
- The TransactionEncoder is imported from the mlxtend.preprocessing module was used to transform the list of itemsets into one-hot encoded Boolean matrix.
- Each row represents a transaction, and each column represents an item.
 - True (1) indicates that the item was present in the transaction.
 - False (0) indicates that the item was not present in the transaction.

2) Application of the Apriori Algorithm

- Apriori Algorithm was used for the low basket size problem because this algorithm helps in identifying the patterns that how items are related which is useful for market basket analysis.
- From the mlxtend.frequent_patterns library apriori function was applied to the one-hot encoded dataset with the minimum support defined.
- frequent_itemsets were generated and then passed to association_rules() which is imported from the library, this step is done to calculate the following metrics:
 - Support: This metrics measures the ratio of how frequent the item appears in a dataset to total number of transactions.

$$\text{support}(I) = \frac{\text{Number of transactions containing } I}{\text{Total number of transactions}}$$

- Confidence: The probability that, when Item X was purchased, Item Y will also be purchased is known as confidence. It is computed by dividing the number of transactions containing Item X and Y by the number of transaction containing Item X.

$$\text{confidence}(X \rightarrow Y) = \frac{\text{Number of transactions containing } X \text{ and } Y}{\text{Number of transactions containing } X}$$

- Lift: Lift measures how strongly two items associated. It is defined as the ratio of support of two items occurring together in a transaction to multiplication of support of both items.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Support}(A \text{ and } B)}{\text{Support}(A) * \text{Support}(B)}$$

3) Extraction and Filtrating of Association rules

- Frequent itemset were generated using apriori function with a minimum support of 0.02 that is 2% of total transactions, means itemset should appears in at least 2 out of

100 transaction (For Example: the dataset has 9465 transactions so, $0.02 \times 9465 = 189.3$ according to that the itemset should appear in at least 190 out of 9465 transactions).

- The association rules were also extracted using `association_rules()` with metrics confidence greater than equal to 0.1 and lift greater than equal to 0.5.
- Reason behind setting this value for these parameters is that it ensures only those itemset with more than 2% of whole transaction to filter out rare combinations.
- A confidence threshold of 0.1 means 10% is good with the medium dataset with less than 10K rows where stronger rules might be too rare, that's why confidence is kept at 0.1.
- The lift threshold of 0.5 was chosen to include associations that are not entirely random but may still provide actionable insights.
- So talking about few more terms that is necessary to know in order to understand the work done in the project.
 - **Antecedents:** In association rules, the antecedents represents the items that are already purchased in a transaction.
 - **Consequent:** It refers to an item or a set of items that is likely to get purchased after antecedent.

Problem 2: Sales Variability and Operational Inefficiency

For the problems like uneven sales or demand, forecasting methods were chosen based on their ability to build time-based patterns so that to analyse historical sales pattern and identify or predict future demand patterns. SARIMAX was selected for its effectiveness in handling structured seasonal data with autoregressive components. Other than that XGBoost, a powerful machine learning algorithm, it has ability to train on non-linear data using calendar variables. Facebook's prophet was also used added to capture trends and seasonal effects in a business. After that, Monte Carlo simulation was used to estimate uncertainty in model. Together, these models addressed the business problem by improving demand predictability, therefore enabling better resource planning and inventory control.

Following are the steps taken in doing market basket analysis.

1) Aggregating Daily Sales

- Transactional data is grouped on basis of date to calculate the total items sold daily.
- This step transformed the dataset into a time series format required for forecasting.

2) Time Series Decomposition

- The sales then decomposed into trend, seasonality, and residuals (to calculate the errors).
- This helped visualize weekly patterns and irregular demand.

3) SARIMAX Forecasting

- SARIMAX was applied to model the time series using bakery's historical sales data.
- Parameters $(p,d,q) = (1,1,1)$ and seasonal order $= (P,D,Q,s) = (1,1,1,7)$ were selected using AIC (Akaike Information Criterion), a metric that balances model accuracy and complexity – lower AIC means better model.
- These model captured structurally weekly trends.

4) XGBoost Regression Forecasting

- XGBoost was trained using independent features such as:
 - Date -- Date of Transaction
 - Month – Month of Transaction
 - Weekday – Which day of the week (example- 0: Sunday, 1: Monday, 2: Tuesday ... so on)
 - Dayofyear – which day of the year in number
- Sales column (feature) is used as target column.
- Parameters of XGBoost are:
 - n_estimator: 100
 - max_depth: 3
 - learning_rate: 0.1
 - random_state: 42
- This was done to capture complex sales pattern.

5) Prophet – A Facebook Model

- Prophet is used to measure trend and seasonality in additive manner.
- It handles holiday effects and any abrupt trend changes.
- Forecast made by the model were visualized with upper and lower bounds, making easy to interpret.

6) Monte Carlo Simulation for Uncertainty

- Monte Carlo Simulation was used to run multiple demand scenario based on historical variations.
- It produced probabilistic ranges of possible sales, supporting safety stock and staffing decisions under uncertainty.

7) Model Evaluation

- There are various evaluation metrics used for analysing how well the models used for problem solving mentioned in the report are performed
 - Mean Absolute Error (MAE): It measures the average absolute difference between predicted and actual sales.
 - Root Mean Squared Error (RMSE): It Penalized larger errors more heavily, this indicates the model is robust or not.
 - R-Squared (Coefficient of Determination): It is used for XGBoost to quantify how well the model explains variability in sales, more R-square the better model performed.

Results and Findings

This section contains key findings derived from analysis of the bakery's transactional data. The findings are primarily presented in a problem-wise manner. However, certain general visualization are also included to provide some insights of business.

1. Top 25 most frequently sold items in the bakery

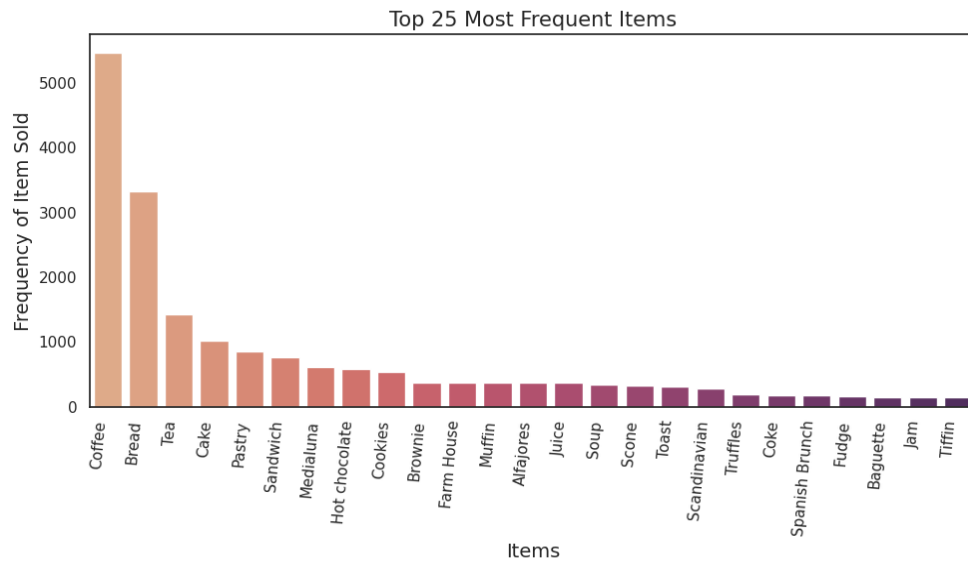


Figure 1.1

Figure 1.1 shows the individual item sold from 30-october-2016 to 9-April-2017. It shows what the most popular items in the bakery are.

Key Observations:

- The bar chart displays the 25 most sold items based on total frequency across all transactions.
- Coffee is the most frequency purchased item, followed by Bread, Tea, Cake, and Pastry.
- There is a steep drop in frequency after the top two items (Ex: Coffee and Bread), indicates that most sales are concentrated around few popular products.
- This pattern indicates that customer often make purchases around popular items, this validates the problem of low basket size.
- These high selling items (Ex: Coffee and Bread) can serve as anchor product for designing combo deals and cross-selling strategies to increase revenue per transaction.

2. Basket Distribution Across Transactions

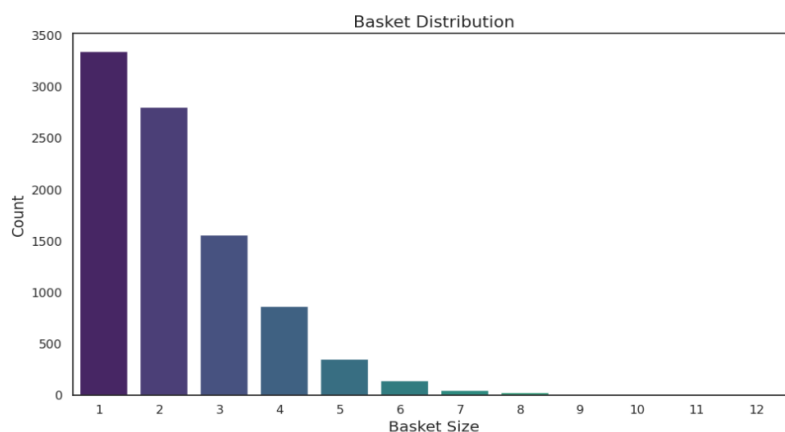


Figure 1.2

Key Observations:

- The charts displays the frequency distribution of basket size, representing how many items customers bought per transactions.
- Most transactions contains only 1 or 2 items, this indicates a dominant single-item purchase behaviour.
- There are 3352 transactions which contains only 1 item in their Basket, while 2803 transaction with 2 items in basket, and 1560 transactions with 3 items, and so on.
- There are very less Basket with more than 3 or 4 items and those with more than 6 are very rare.
- This Basket Size distribution is highly skewed towards low basket size, indicates that this chart is valuable for first problem.

3. Product Association based on Lift (Apriori Rules)

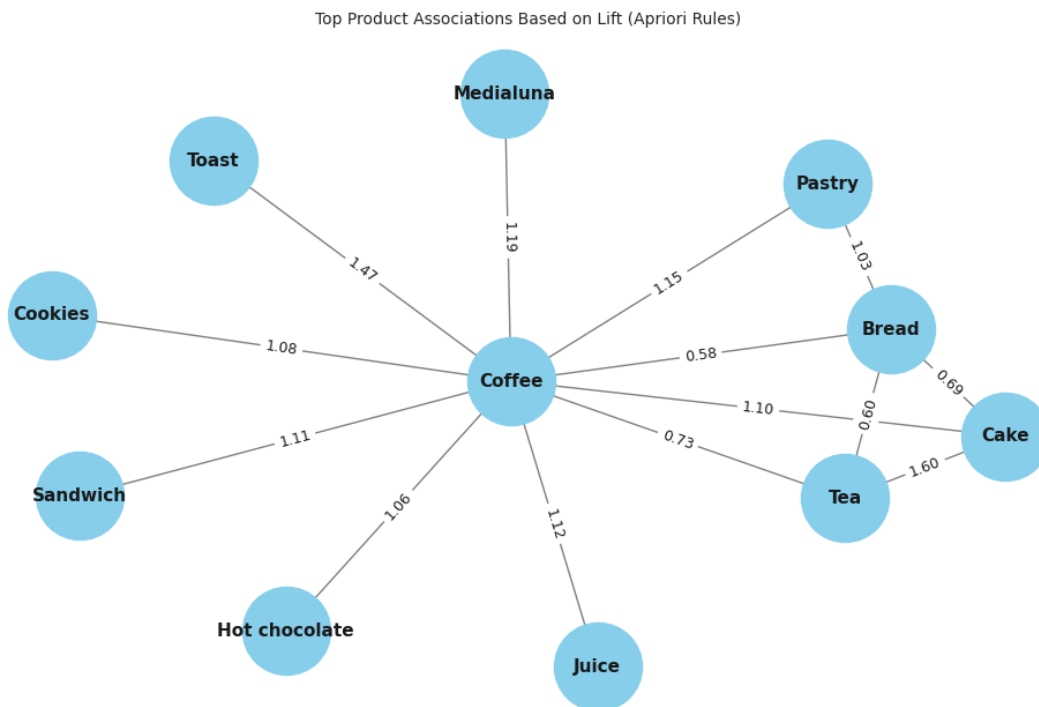


Figure 1.3

After Conduction Market Basket Analysis, Some steps were taken in order to solve the first problem and step was Association rule mining, and Apriori Algorithm was used, in figure 1.3, nodes are the items and edges are connection between items and number of those edges is lift, more the lift the higher chances of those combination of items being bought together. Here, Coffee is the centre node because coffee is the top selling item in the bakery and it will be used to identify connection between items.

Key Observations:

- The network graph visualizes product associations based on Lift metrics from the Apriori Algorithm, highlighting these items have higher chances to be purchased together than by chance.
- Strong Associations were observed between coffee and items like Toast (Lift: 1.47), Medialuna (Lift: 1.19), Pastry (Lift: 1.15), and Cake (Lift: 1.10).
- A few notable associations also exist between other items such as, Bread-Cake (Lift: 0.69), suggests potential bundling of baked goods, Tea-Cake (Lift: 1.60) and Bread-Tea (Lift: 0.60), highlights opportunities for pairing beverages with baked items beyond just coffee.
- This multi-directional graph confirms that bundling strategies can be designed not just around coffee but also for other popular or complementary items such as Tea for encouraging customer to buy more than 1 or 2 items per transactions.

4. Weekly Items Sales Chart

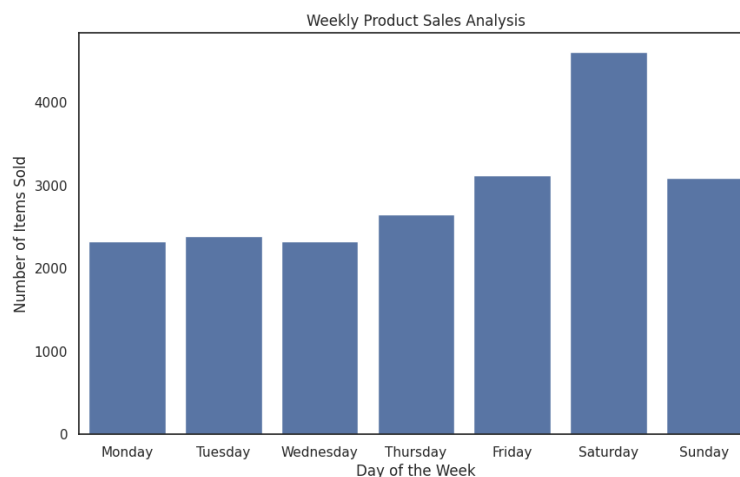


Figure 2.1

Figure 2.1 shows weekly product sales pattern, this bar chart visualizes the number of items sold throughout the week to identify short-term sales patterns, which addresses problem 2: Sales variability and operational inefficiency due to inconsistency in demand.

Key Observations:

- Saturday has the highest sales among other days, followed by Friday and Sunday, indicating peak customer engagement.
- Monday to Wednesday reflects a drop in demand with relatively lower sales activity, suggesting a quiet start of week after a good weekend.

- Friday and Sunday demonstrates moderately high sales, as compared to Monday to Wednesday, acting as transition period where demand begins to increase.
- This pattern highlights a consistent cyclical weekly demand structure, which is critical for resource planning.

5. Heatmap of Transactions by Week and Hours

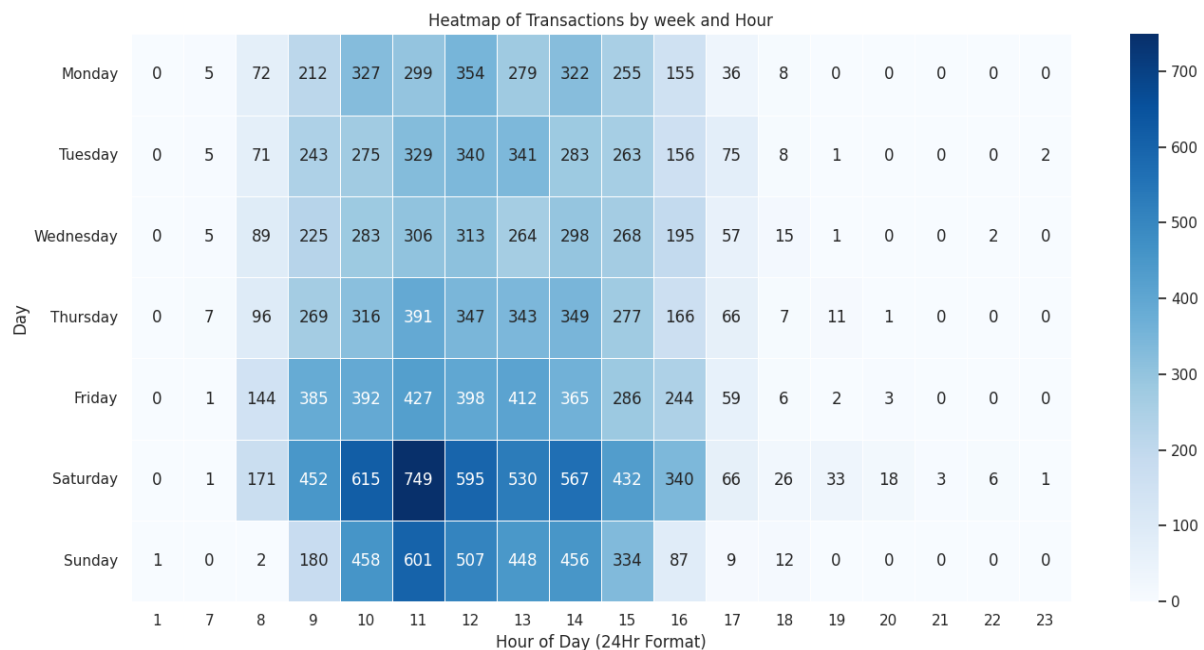


Figure 2.2

Figure 2.2 displays the volume of transactions across each day of the week and hour of the day, helping identify temporal spikes in demand. X-axis is the Hour of the day and Y-axis is the day of week indicating the number items sold at what part of time and day.

Key Observations:

- This heatmap of the Bakery data tells the operational hours of the bakery so, the bakery starts its operations at 7:00 am at morning and closes at 1:00 am at night.
- The period of day that was in dataset as column named “period_day” is divided into 4 sections: 7:00 am to 11:59 am (Morning), 12:00 pm to 16:59 pm (Afternoon), 17:00 pm to 20:59 pm (Evening), and 21:00 pm to 1:00 am (Night).
- As the heatmap highlights that on Saturday between 10 AM and 2 PM sees the highest transaction volume, followed by Sunday and Friday same timeline.
- Weekdays including specially Monday, Tuesday, Wednesday, and Thursday are more evenly distributed but lower activity was recorded, which is concentrated between 9 AM to 2PM.

- One of major things to focus is that there is hardly any sales after afternoon especially in evening and night, this was expected as the bakery is based in Scotland and the dataset was recorded from October till April during winter months. Despite low demand, the bakery remained open late till 1:00 am, indicating inefficient resource utilization during non-peak hours.

6. SARIMAX Forecast – Daily Sales

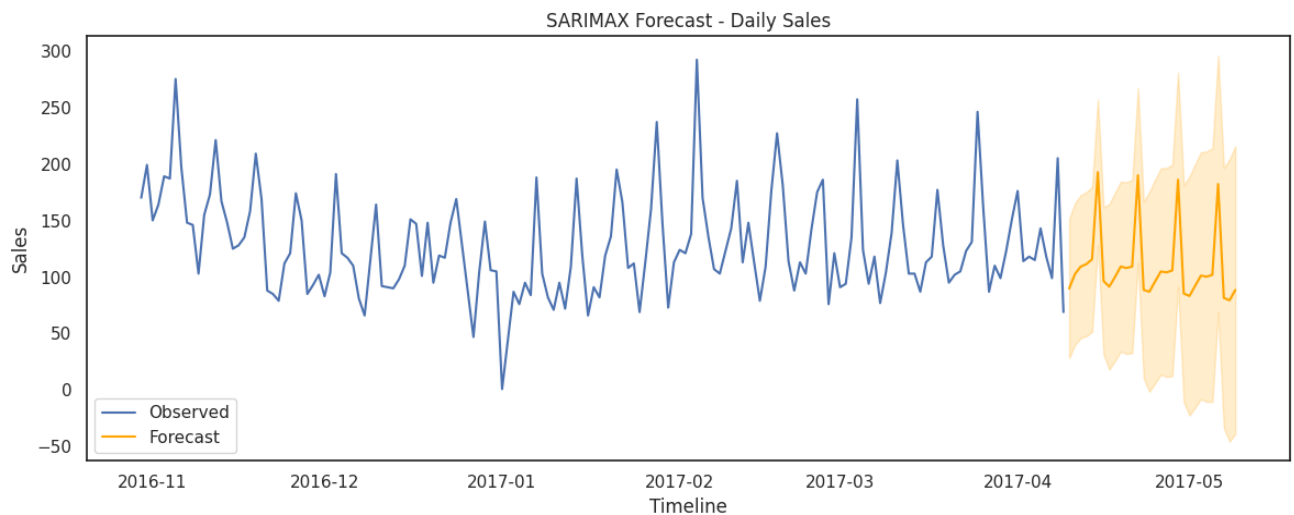


Figure 2.3

Figure 2.3 line chart illustrates the Observed which is the actual daily sales and forecasted daily sales using the SARIMAX (Seasonal Autoregressive Integrated Moving Average with Exogenous Variable) model. This model was selected for its ability to handle seasonal fluctuations, autocorrelations, and trend components in time-series data.

Key Observations:

- The observed (Actual) sales data shows high daily fluctuation with visible weekly spikes, reinforcing the presence of sales seasonality.
- The forecasted values (with confidence intervals) captures this periodic nature, indicating recurring weekly demand surges, typically on weekends.
- The forecast also reflects uncertainty bands that is widening over time suggesting a need for dynamic planning and regular model re-training on further daily and weekly sales.
- The SARIMAX was chosen over ARIMA because its capability to incorporate seasonality, which aligns well with the bakery's sales behaviour.

7. XGBoost Forecast vs Actual Sales (30-Day Window)

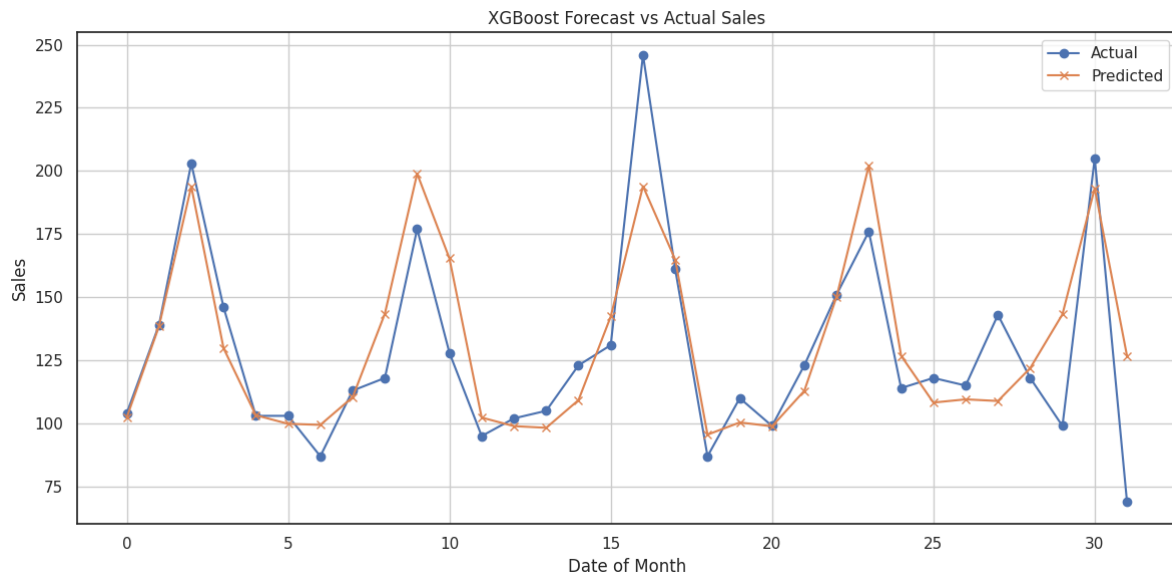


Figure 2.4

Figure 2.4 shows the line chart which compares the predicted sales values from an XGBoost regression model to the actual observed sales over a 30-day period. The model achieved a Root Mean Squared Error (RMSE) of 20.95 and R-Squared of 0.697, indicates a strong fit. XGBoost is a powerful machine learning algorithm well-suited for captured non-linear patterns and interactions in time series data. XGBoost was trained the aggregated data which is grouped by date so there is date time column as index and column named “Sales” which was the sales of that day.

Key Observations:

- The XGBoost model accurately tracks the recurring spikes in demand, particularly near weekends, aligning with weekly sales seasonality.
- Some deviation are observed, especially during mid-month peaks, where actual sales exceeded predictions, it is likely due to external events not captured in model features.
- Despite minor underestimations at extreme points, the overall sales trend and cycle are well captured, indicating model reliability for operational planning.

8. Sales Forecasting using Prophet – A Facebook Model

Below Figure is also a line plot that displays sales forecast generated using prophet time series model over a 30 day period. The model achieved a Root Mean Squared Error (RMSE) of 28.41 and R-Squared of 0.60, indicating a moderately strong fit for capturing underlying sales patterns. Prophet is particularly suitable for time series data with clear seasonality and trend structures.

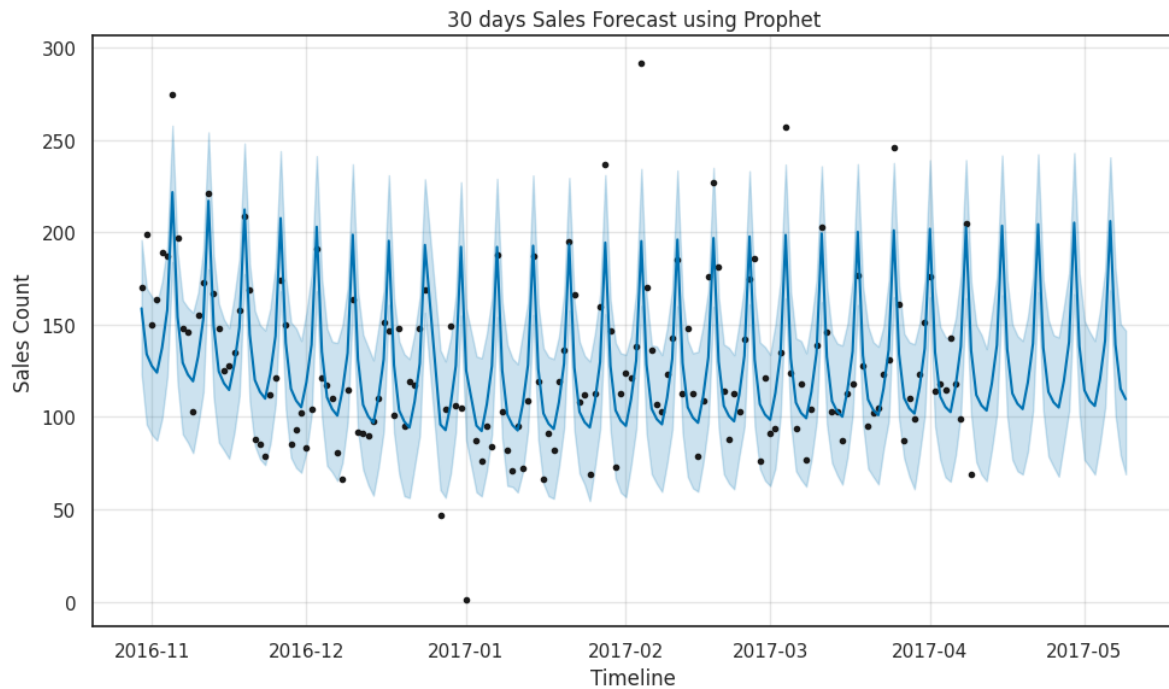


Figure 2.5

Key Observations:

- The black dots in the charts are the actual historical sales, the blue line shows the predicted values, and the shaded region indicates the model's confidence interval of 95% means that the actual sales will fall within the shaded region.
- The forecast line aligns well the historical trend, while uncertainty interval widens towards the end of horizon, reflecting increasing forecast uncertainty.
- The model provides interpretable and smooth predictions, suitable for operational level planning.

9. Monte Carlo Simulation of Daily Demand for next 30 days

Below Figure visualizes the results of Monte Carlo simulation applied to forecast daily done by SARIMAX shown in Figure 2.3, this simulation is done on daily sales demand over a 30-day horizon. The grey lines represent individual simulation runs, each showing a potential sales outcome based on random sampling from historical sales patterns. The blue line indicates the average of all simulations, reflecting the expected demand under uncertainty,

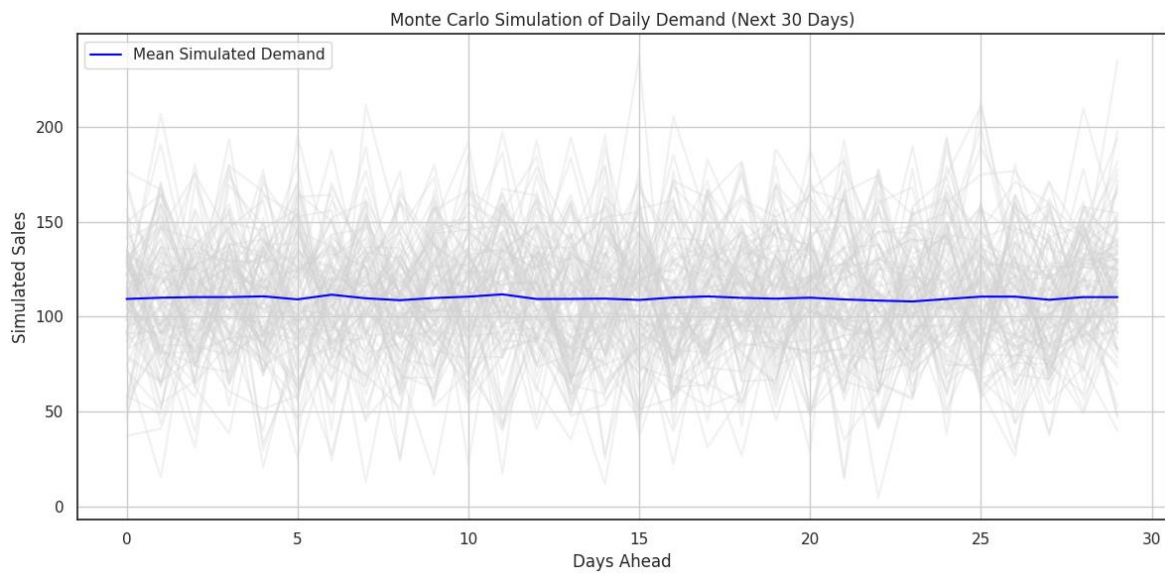


Figure 2.6

Key Observations:

- The mean simulated demand remains fairly stable (~ 105-110 daily sales), which aligns with historical averages.
- The spread of simulation paths shows significant day-to-day variability, emphasizing the uncertainty and potential volatility in short-term demand.
- This range includes possible extreme low or high sales, which the deterministic models may not capture.

Interpretation of Results and Recommendations

Interpretation of Results

The charts that are provided in the previous section were produced by doing data analysis on the bakery's sales data uncovered meaningful patterns related to customer purchasing behaviour and operational performance. By examining transaction-level trends, item associations, and forecasting outputs, the study provides clear evidence of inefficiencies and growth opportunities. The following interpretations connects these analytical findings to the business problems identified earlier, highlighting how data-driven insights can guide strategic improvements.

Problem 1: Low Average Basket Size

- The analysis shown in Figure 1.2 revealed that a majority of transactions consists of single-item purchases, which directly supports the business problem of low average basket size. The insights are reinforced by the frequency distribution of items and supported through market basket analysis conducted by Apriori Algorithm. The rules generated by association rule mining showed high lift values between commonly purchased pairs such as Coffee with Toast, Medialuna, Pasty, Cake, indicates strong co-occurrence patterns.
- The network graph of item associations shown in Figure 1.3 further highlighted coffee as a central product node, consistently linked with multiple high-frequency items. The visual evidence confirms the presence of natural product bundling behavior, even though it is not currently leveraged operationally. The dataset's consistency weekday behavior shown in Figure 2.1, with similar sales volumes and single item dominance shown in Figure 1.1, sales of Coffee is very high as compared to other items, points to a routine buying pattern among customers – potentially driven by time constraints or lack of promotion for additional items.
- These findings establish a strong analytical foundation for identifying missed opportunities. The graph (Figure 1.3) tells more than just showing item popularity – they illustrate how the lack of cross-selling, bundling, or targeted promotions is a key factor behind low basket size (Figure 1.2) leading to low revenue per transaction. Therefore, the observed associations present a compelling case for developing data-driven strategies aimed to increasing the number of items sold per transactions.

Problem 2: Sales Variability and Operational inefficiency

- The Weekly sales analysis shown in Figure 2.1 and Daily and Hourly sales analysis shown in Figure 2.2 through heatmap, uncovered significant inconsistency caused by unpredictable demand that directly aligns with the problem. The weekly sales chart (Figure 2.1) showed a clear concentration of customer activity on Fridays, Saturday, and Sundays. While the hourly heatmap revealed that the bulk of transactions occurred between 9 AM to 2 PM, particularly on weekends that can be seen in Figure 2.2. In contrast, weekday evenings consistently recorded minimal sales, despite the bakery remained operational during low-demand hours.
- This imbalance in demand is further reinforced by the outputs of SARIMAX (Figure 2.3) and Prophet model (Figure 2.5), both of which captured reporting weekly cycles and predictable weekend spikes. These time series forecasts confirmed that the bakery experiences structured demand variability that can be anticipated with appropriate modelling. The XGBoost model (Figure 2.4) demonstrated high accuracy in short-term predictions, aligning closely with observed sales values, especially during high-traffic days. Additionally, the Monte Carlo simulation (Figure 2.6) exposed the degree

of uncertainty in demand, even when the average forecast remained stable. The wide distribution of possible sales scenarios emphasized the risk of both over-preparation and under-preparation.

- These insights shows that while demand patterns are not entirely random, the bakery currently lacks a data-driven operational framework to manage them effectively. The interpretation of these results suggests that improvements in forecast-informed staffing, time-bound production, and inventory planning could significantly reduce resource waste and improve cost-efficiency.

Recommendations

Problem 1: Low Average Basket Size

The analysis confirms that most customer purchase only a single item per transaction, even though product associations suggest a strong tendency for certain items to be bought together. The following recommendations are grounded in data insights from associations rules, item frequency analysis, and weekday transaction pattern:

1. Launch Data-Driven Combo Offers

- Based on high lift values observed in Figure 1.3 between Coffee and items like Toast, Medialuna, Cake, Pastry, Juice, Hot Chocolate, and Cake. These are the one of highest recorded combinations as using lift metrics.
- So, various combinations can be made such as;
 - Coffee + Cake, Coffee + Pastry, Coffee + Bread Combo
 - Tea + Cake, Tea + Bread
 - Bread + Pastry
- Price each combo slightly below the combined price of individual items to encourage multi-item purchases.
- Schedule these items combo offers primarily during morning hours, when beverages purchase peaks.

2. Introduce POS-based Placement and Signage

- Incorporate scripted prompts into point-of-sale process or digital kiosk interface.
- Examples are:
 - When a customer orders a Coffee, prompt staff to ask: “Would you like to add a slice of Cake for just £0.50 more”.

- This will leverage association rules in real-time and nudges customer towards multi-item purchases.

3. Optimize Product Placement and Signage

- Rearrange store layout to co-locate frequently associated products. For Example:
 - Displaying pastries besides Coffee dispensers
 - Use Signage like “Best with Coffee” or “Perfect Pairing to start your day with”.
- This will draw attention towards suggested item combinations.
- This aligns with patterns observed in the network graph (Figure 1.3) and increase visual triggers for add-on purchases.

4. Implement Weekly Loyalty Discounts for Multi-Item Orders

- To improve weekday sales and average order value, offer loyalty rewards or discount codes that activates only on weekday multi-item purchases. For example:
 - Buy 2 items, Get 10% Off – Monday to Thursday Only
- This uses consistent weekday traffic to drive more value per transactions, especially during slow sales periods.

Problem 2: Sales Variability and Operational Inefficiencies

Sales Pattern exhibits clear variability across days and times, with peak transactions occurring in weekend mornings and sharp drops during weekdays evening as shown in heatmap (Figure 2.2). Forecasting models confirms predictable demand cycles, while simulation highlighted uncertainty. The following recommendations aim to align staffing, inventory, and operational hours more closely with actual demand behavior.

1. Implement Forecast-Based Inventory Planning

- Use Prophet for weekly stock planning to accommodate recurring weekend surges. Pair it with XGBoost for short-term adjustments. Example:
 - Day-before demand tuning
- This ensures adequate stock on peak days (Friday-Sunday) while minimizing wastage during low-demand periods (Monday-Wednesday), as supported by both forecasting model shown in Figure 2.4 and Figure 2.5.

2. Realign Staff Scheduling with Hourly Demand

- Based on the hourly heatmap (Figure 2.2), schedule more staff during 9AM - 2PM on weekends, when sales activity is highest.
- Reduce Staff during weekday evenings, where transaction volumes are consistently low.
- This ensures cost-effective staff allocation without compromising service quality during busy hours.

3. Use Monte Carlo Simulation for Buffer Stock Decisions

- The Monte Carlo Simulation shown in Figure 2.6 showed that while average demand is stable (~105-110 daily sales), the range of possible outcomes varies widely.
- Using the upper 95th percentile from the simulation to define buffer stock thresholds, ensuring readiness for occasional spikes without overproducing.

4. Adjust Operating Hours During Off-Peak Seasons

- Sales data between October and April (winter months) showed negligible transactions post 5 PM. Consider closing the Store early on weekdays during these months to reduce utility and staffing costs.
- This aligns operational hours with actual customer behavior and seasonal demand patterns.

5. Launch Off-Peak Incentives to Smooth Demand

- To reduce over-reliance on weekend peaks, launch evening deals on weekdays. These targeted promotions can help redistribute demand, improve inventory flow, and make better use of underutilized staff during quieter time blocks